UNOCHA GEO-INSIGHT HACKATHON · MAY 2026

GEO-INSIGHT

Which Crises Are Most Overlooked?

An agentic command center surfacing mismatches between humanitarian need and funding coverage — across 77 active crises worldwide.

SUBMITTED BY

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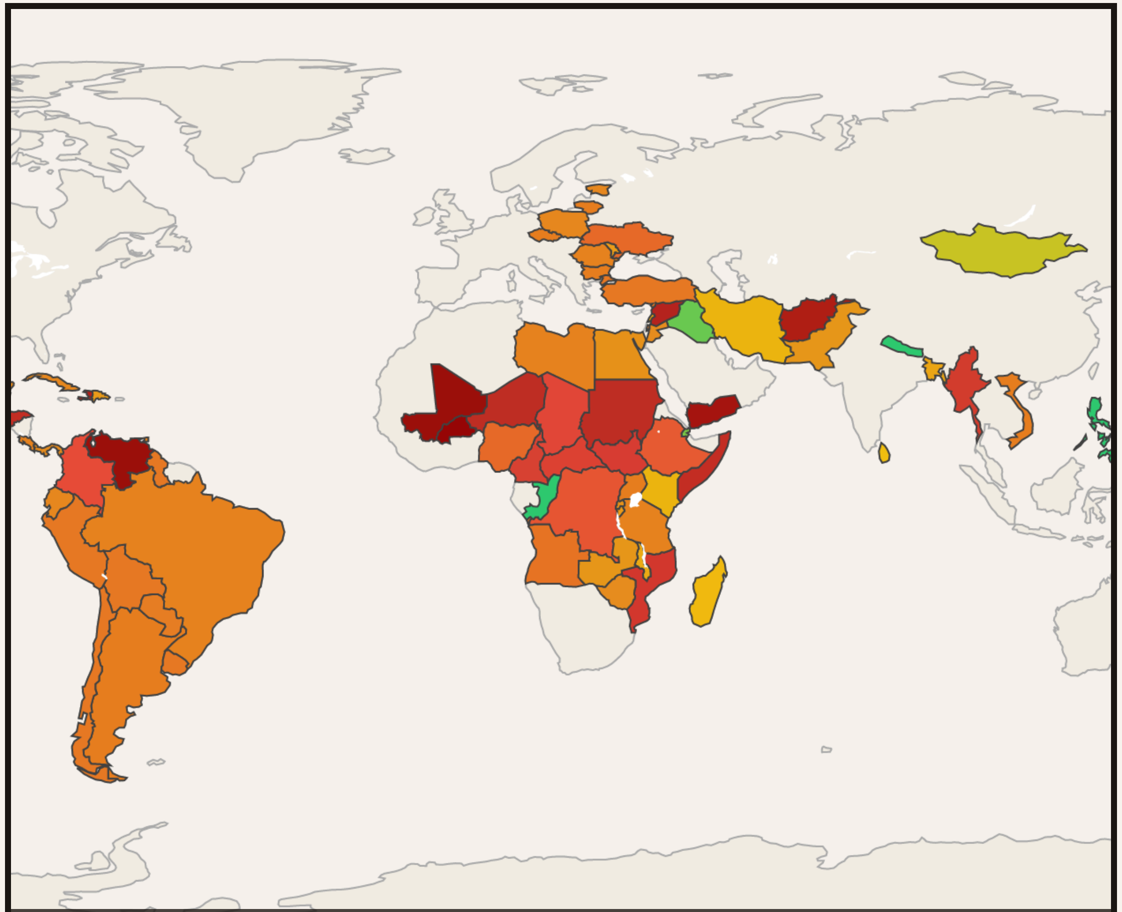
CMU AIM · Solo Build · May 18–21, 2026



GAP SCORE WORLD MAP · RED = MOST OVERLOOKED · 77 COUNTRIES

77 COUNTRIES	**229M** PEOPLE IN NEED	**8** STRUCTURAL NEGLECT	**6** AGENT TOOLS

CLAUDE CLAUDE-SONNET-4-6 · MEDALLION PIPELINE · MLFLOW TRACING · CI · HDX · FTS · INFORM · RESPONSIBLE AI SCORECARD

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GEO-INSIGHT

- PROBLEM
- ARCHITECTURE
- GAP SCORING
- DATA SOURCES
- DEMO
- LIMITATIONS
- WORKFLOW FIT

Which Crises Are *Most Overlooked?*

Surfacing mismatches between humanitarian need and funding coverage

THE PROBLEM

Humanitarian data mixes two very different signals: **objective severity** (scale of need) and **funding coverage** (what's actually resourced). Without a system to combine them, critical gaps go unnoticed.

EXAMPLE QUERIES COORDINATORS ASK

- "Which crises have the most people in need but lowest fund allocations?"
- "Are there countries where funding received is under 10%?"
- "Which regions are consistently underfunded across 3+ years?"

COUNTRIES TRACKED

77

Active crises, 2 data tiers

PEOPLE IN NEED

229M

HNO-tier countries

AVG. COVERAGE

38%

Funding received / requested

STRUCTURAL NEGLECT

8

3+ consecutive underfunded years

CLAUDE REACT AGENT | MEDALLION PIPELINE | MLFLOW TRACING | RESPONSIBLE AI SCORECARD

> *"The goal is tools that help decision-makers ask better questions — not tools that make choices for them."*
>
> — UNOCHA GEO-INSIGHT CHALLENGE BRIEF

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End-to-End *Architecture*

Medallion pipeline · Claude ReAct agent · Streamlit dashboard

HDX HAPI HNO · HRP · INFORM → **FTS API** FUNDING FLOWS · PLANS → **BRONZE** RAW DELTA TABLES → **SILVER** 77 COUNTRIES · 2 TIERS → **GOLD** GAP SCORES · RANKED → **CLAUDE AGENT** 6 TOOLS · REACT → **STREAMLIT UI** 6 TABS · CHAT · MAP

TWO-TIER DATA MODEL

TIER	COUNTRIES	SIGNALS
HNO	24	Full gap score · PIN · sector breakdown
FTS PROXY	53	Funding gap + severity · PIN = N/A

AGENT TOOLS (6)

- `decompose_query` · NL → structured filters
- `query_gold_table` · parameterized SQL
- `semantic_search` · embedding retrieval
- `generate_briefing_notes` · per-crisis memo
- `validate_ranking` · sanity checks
- `red_team_challenge` · counter-arguments

KEY DESIGN CHOICES

Medallion architecture — Bronze → Silver → Gold separation lets scoring logic evolve without re-ingesting raw data

MLflow tracing — every agent query traced with root span + per-tool child spans; auditable by judges

Hybrid scoring — structured ratios (funding gap) + contextual signals (INFORM, neglect, donor HHI)

TECH STACK

CLAUDE CLAUDE-SONNET-4-6 · STREAMLIT · MLFLOW · DELTA LAKE · PLOTLY · PANDAS · HDX HAPI · FTS API

Defensible *Gap Score*

Every component has an explicit humanitarian rationale

```
# 1. Funding gap (primary signal)
funding_gap = 1 - (received / requested)
# 2. Need scale: log-normalize vs global max PIN
need_scale = log1p(PIN) / log1p(max_PIN)
# 3. INFORM severity multiplier (0-10 → 0-1)
sev_mult = INFORM_score / 10.0
# 4. Composite base score
base = gap × (0.435 + 0.348×scale + 0.217×sev)
# 5. Structural neglect multiplier
gap_score = base × (1 + yrs_underfunded × 0.15)
```

WHY EACH COMPONENT

COMPONENT	WHY IT'S HERE
Funding gap	Directly answers the core question — what fraction of need is unresourced
Need scale	Prevents a tiny crisis at 0% coverage outranking a 50M-person crisis at 10%
INFORM severity	Independent validation — not self-reported; avoids gaming of HNO figures
Neglect factor	Separates structural chronic neglect from acute one-time emergencies
Donor HHI	Single-donor reliance = invisible systemic risk; bonus task requirement

BONUS: DONOR CONCENTRATION

Herfindahl-Hirschman Index (HHI) computed from FTS Flows API for 72 countries. A crisis with HHI > 0.25 or top-donor share > 40% receives a **concentration risk flag** — single-donor dependency = hidden fragility not visible in coverage ratio alone.

Haiti HHI = 0.307 ⚠ | Venezuela top-1 = 45.7%

UNCERTAINTY QUANTIFICATION

Monte Carlo simulation (1,000 runs) with PIN ±30% and funding ±20% triangular distributions → **P10 / P50 / P90** confidence bands shown per country. Pareto-efficient crises flagged as **★ frontier**.

Data Sources & *Messy Data* Handling

What we used, what we worked around, and how we stayed honest

SOURCE	WHAT IT PROVIDES	CHALLENGE HANDLED	STATUS
HDX HNO	People in need, sector breakdown, severity	Only 24 countries → HNO tier	FRESH
FTS API	Funding requested / received / pledged	Proxy for 53 more countries, no PIN	2025–26
INFORM Severity	Independent crisis severity (0–10)	Default 5.0 when missing	2025
CBPF Pooled Funds	Pooled fund allocations by country	Only 2018 vintage available	STALE ⚠
FTS Flows API	Donor-level flows → HHI, top donor share	72 / 77 countries covered	PARTIAL

MISSING DATA PROTOCOL

- No HRP → coverage = 0%, gap score × 0.5 penalty (less certain)
- Missing PIN → INFORM severity used as proxy; FTS proxy badge shown
- Stale data → staleness badge (🕐) in UI; days since update shown
- Requested vs pledged vs received → always use **received**; all three shown in detail

DECLARED EXTERNAL MODELS/APIS

Claude claude-sonnet-4-6 (Anthropic)
Query decomposition + briefing notes

INFORM Global Crisis Severity Index
Independent severity validation

HDX HAPI REST API + FTS api.hpc.tools
Primary data access layer

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System in *Action*

3 query examples · 6-tab dashboard · conversational agent

01 GLOBAL RANKED TABLE

RANKED TABLE | WORLD MAP | ASK AGENT

RANKED CRISES BY GAP SCORE

#	Country	Gap Score [CI]	Coverage	PIN	Neglect
1	Burkina Faso	1.165	13.8%	4.40M	structural
2	Mali	1.136	16.0%	5.10M	structural
3	Venezuela	1.132	15.4%	7.90M	structural
4	Yemen	1.108	13.1%	23.10M	ongoing ★
5	Afghanistan	1.081	14.9%	21.89M	ongoing
6	Haiti	1.065	22.5%	6.41M	structural

77 countries · HNO ✓ = full gap score · Monte Carlo P10/P50/P90 shown

02 AGENT: CBPF QUERY

Which structurally neglected crises in Sub-Saharan Africa should a CBPF fund manager prioritize this year?

CBPF Priority Assessment: Structurally Neglected Crises — Sub-Saharan Africa (2025)

⚠ Data Confidence: MEDIUM — gap scores are directional signals, not precise measurements

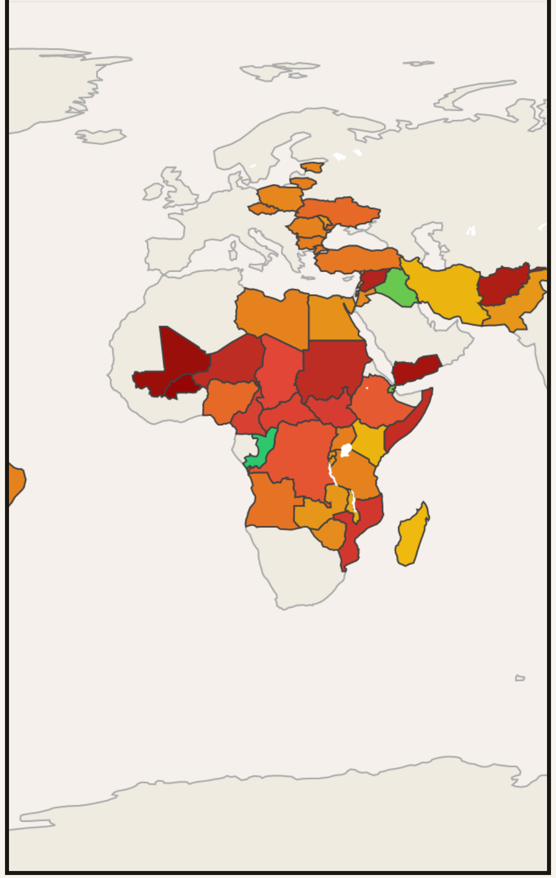
#	Country	Gap Score	Coverage	PIN	INFORM
1	Mali	1.025	24.6%	6.4M	8.3/10
2	Mozambique	0.986	23.3%	1.3M	7.5/10
3	Cameroon	0.951	23.0%	3.3M	5.0/10
4	Burkina Faso	0.894	34.2%	5.9M	8.4/10

Mali #1: 75-cent shortfall on every dollar requested. Access denial in Ménaka/Kidal may affect ~30% of PIN — fund managers should verify accessible zones before treating gap as fully actionable.

onsecutive underfunded years) · All have

03 CHOROPLETH MAP

Gap Score World Map · 77 countries · higher = more overlooked



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What This System *Does Not Know*

Limitations · Responsible AI Scorecard · 7 dimensions

KNOWN LIMITATIONS

- **FTS proxy PIN = 0** — 53 countries scored on funding gap + severity only; gap score underestimates true need
- **CBPF data is 2018 vintage** — pooled fund allocations flagged as stale; FTS used as primary funding source
- **Coverage ≠ access** — 0% funding may mean access is blocked, not just neglected; system cannot distinguish
- **No last-mile tracking** — system cannot verify if aid reached beneficiaries; funding received ≠ impact
- Sector-level analysis available for HNO-tier countries only (24 of 77)

RESPONSIBLE AI SCORECARD

DIMENSION	IMPLEMENTATION	
Transparency	Formula fully documented, open weights, all sources declared	✓
Uncertainty	Monte Carlo P10/P50/P90 bands on every score	✓
Fairness	Two-tier badge prevents false equivalence; proxy countries flagged	✓
Neutral Framing	"May warrant" language; no automated decisions; human in loop	✓
Human Oversight	All outputs are recommendations; red_team_challenge tool built in	✓
Data Governance	All sources public; data_sources.md declared; CBPF staleness flagged	~
Model Governance	MLflow traces every query; parameters logged; reproducible	✓

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How This Fits *Real Workflows*

Who uses it · what they do next · what could go wrong

USER ROLES

HC Humanitarian Coordinator — morning briefing query → triage agenda → field cross-check

DA Donor Advisor — "where should next $10M go?" → defensible shortlist → donor contact memo

OA OCHA Analyst — sector gap drilldown → CAP process flag → advocacy brief

AFTER SEEING RESULTS, NEXT STEPS

1. Cross-check with country team / field reports — system is decision support, not decision-maker
2. Flag structural neglect countries for Consolidated Appeals Process (CAP)
3. Share agent-generated briefing note with donor contacts or media

WHAT COULD GO WRONG

Overreliance: gap score ≠ absorptive capacity — 0% coverage may mean access is blocked, not neglected

Gaming: if rankings become public, actors could inflate HNO PIN estimates to attract funding

Staleness: data lags mean a newly erupting crisis may score low until next HNO cycle

"Good humanitarian data work does not just optimize numbers. It respects context."

DELIVERABLES SUBMITTED

✓ WORKING PROTOTYPE ✓ TECHNICAL WRITE-UP ✓ 15-MIN DEMO VIDEO ✓ GAP SCORING LOGIC ✓ FAILURE CASES ✓ BONUS: STRUCTURAL NEGLECT ✓ BONUS: DONOR CONCENTRATION